

Supplement:

Engineering What an Optical Sensor Cannot See

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This Supplement is an audit trail, not an alternate manuscript. It contains the full proofs of the four main-text results (S1–S4), the model and protocol capsule with engines, seeds, and commit hashes (S6), the theorem-consistent honest negatives (S5), and the frozen three-ledger scaling-gate (TLSG) table (S4.3). Every number traces to a single committed frozen artifact recorded in `paper2_optica/CLAIM_SOURCE_MATRIX.md`; no datum was generated for this document. Equation, section, table, and figure labels carry an “S” prefix; bare references point to the main article.

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S1 Statistical Noether wall: proof and periodic table

Let a scene x lie on a smooth parameter manifold $\mathcal{X}$; let the complete observed record have a dominated law P_x with density $p(y \mid x)$; and let a Lie group G act smoothly on $\mathcal{X}$ with infinitesimal generator $X_\xi(x)$ for ξ in the Lie algebra.

Theorem 1 (Statistical Noether wall). *If $P_{g \cdot x} = P_x$ for every $g \in G$, then for every algebra element ξ*

$$D_x \log p(Y \mid x)[X_\xi(x)] = 0 \quad a.s., \quad I_x X_\xi(x) = 0. \tag{S1}$$

Consequently every f -divergence, Chernoff distance, and quotient information jet between two points on a common orbit is zero, and no estimator, moment order, exposure count, or reconstruction prior can distinguish points on that orbit.

Proof. Invariance gives $\log p(y \mid \exp(t\xi) \cdot x) = \log p(y \mid x)$ for all t and almost every y . Differentiating at $t = 0$ yields the directional score identity (S1). Taking the outer product and expectation gives $I_x X_\xi = \mathbb{E}[(D_x \log p[X_\xi])(\cdot)] = 0$. Because the complete laws coincide at every finite orbit displacement $P_{g \cdot x} = P_x$, any f -divergence $\mathcal{D}_f(P_{g \cdot x} \| P_x) = \mathcal{D}_f(P_x \| P_x) = 0$; the same holds for the Chernoff distance and for any nuisance-profiled quotient jet, which is an infimum of such divergences. $\square$

Lemma 1 (Local converse). *Let $\mathcal{D}_x$ be a smooth constant-rank distribution that is integrable and satisfies $D_x \log p(y \mid x)[v] = 0$ for every $v \in \mathcal{D}_x$, almost every y , and every x in a connected neighborhood. Then $p(y \mid x)$ is constant along each integral leaf of $\mathcal{D}$; the leaves are local blindness orbits.*

Table S1. Wall periodic table for an optical chain. The first two rows are the walls confirmed here; the mode-unitary and parity rows are immediate consequences of Theorem 1 (Section S1).

Wall class	Invariance / group	Exact null	Breaking operator
Support	additive translations by $\ker A$	scene modes beyond the transfer range	finite window, aliasing, nonlinear moment
Unitary energy	$E \rightarrow UE$ under total-energy collection	every lossless mode mixing, incl. phase-only	NA clipping, mode-selective collection
Mode-unitary	full U on spatial modes	any unitary mode transform	mode sorter / analyzer
Global phase	$E \rightarrow e^{i\phi} E$ under intensity detection	absolute optical phase	coherent reference
Parity/exchange	$x(r) \rightarrow x(-r)$, symmetric chain	odd tangent modes at a parity-even baseline	asymmetric code or pupil
Polarization-unitary	local/global $SU(2)$ under intensity bucket	polarization rotation at fixed intensity	analyzer, birefringent reference

Proof. By Frobenius integrability the distribution has integral leaves. Along any smooth path $\gamma(t)$ inside a leaf, $\frac{d}{dt} \log p(y | \gamma(t)) = D_x \log p[\dot{\gamma}] = 0$ for almost every y , so $p(y | \cdot)$ is constant on the leaf. Hence, under ordinary smoothness and integrability, continuous local walls are exactly local gauge symmetries of the statistical experiment. $\square$

Remark 1. A symmetry of the underlying wave equation does not by itself create a wall; a wall appears only when the full measurement law inherits the induced parameter action. Discrete degeneracies—parity, conjugation, source–detector exchange—may produce globally indistinguishable pairs without a nonzero infinitesimal tangent; these are walls but not Lie-generated local nulls, and are outside Theorem 1.

The two confirmed walls as group actions. The support wall is the additive group of the unmeasured subspace: for a bandlimited mean operator A , every $h \in \ker A$ gives $x \mapsto x + h$ with $P_{x+h}^{\text{mean}} = P_x^{\text{mean}}$; a hard field pupil restores it to a maximum field leak beyond $2kR$ of 6.6×10^{-16} (KT1, @19338a4). The energy wall is the unitary orbit of a lossless full-aperture bucket $B(E) = \|E\|_2^2$, with phase-only change the diagonal subgroup; the measured null is 2.18×10^{-16} (IT1, @f13a323).

There is no finite universal list independent of apparatus: the complete local list for a declared experiment is the integrable kernel distribution of its score map, and Theorem 1 supplies the classification rule.

S2 Blindness capacity: proof, robustness, and noncomposability

Let $L : \mathcal{C} \rightarrow \mathcal{Y}$ be a calibrated linear leak operator with the code norm whitened by the desired in-band metric, and write singular values ascending $\sigma_1 \leq \dots \leq \sigma_n$. With $\beta_d(L) = \min_{\dim S=d} \max_{c \in S, \|c\|=1} \|Lc\|$:

Theorem 2 (Blindness capacity). $\beta_d(L) = \sigma_d(L)$, attained by the span of the bottom d right singular vectors. For a generalized relative-leak certificate with metric $B^\dagger B$, $\beta_d = \sqrt{\lambda_d}$ with $L^\dagger L v = \lambda B^\dagger B v$.

Proof. Let $v_1, \dots, v_n$ be right singular vectors with singular values $\sigma_1 \leq \dots \leq \sigma_n$ and set $S^* = \text{span}\{v_1, \dots, v_d\}$. For $c \in S^*$ with $\|c\| = 1$, $\|Lc\|^2 = \sum_{i \leq d} \sigma_i^2 |\langle c, v_i \rangle|^2 \leq \sigma_d^2$, so $\beta_d \leq \sigma_d$. Conversely, any d -dimensional S intersects $\text{span}\{v_d, \dots, v_n\}$ (dimension $n - d + 1$) nontrivially, by a dimension count; a unit vector c in the intersection has $\|Lc\|^2 = \sum_{i \geq d} \sigma_i^2 |\langle c, v_i \rangle|^2 \geq \sigma_d^2$, so $\beta_d \geq \sigma_d$. Hence $\beta_d = \sigma_d$. The generalized statement is the same argument for the pencil $(L^\dagger L, B^\dagger B)$ after the congruence $B^\dagger B = R^\dagger R$. $\square$

Theorem 3 (Robust calibration certificate). If a calibration returns $\hat{L}$ with $\|L - \hat{L}\|_{\text{op}} \leq \delta$ and S_d is the bottom- d right singular subspace of $\hat{L}$, then

$$\max_{c \in S_d, \|c\|=1} \|Lc\| \leq \sigma_d(\hat{L}) + \delta. \quad (\text{S2})$$

Proof. For $c \in S_d$ with $\|c\| = 1$, $\|Lc\| \leq \|\hat{L}c\| + \|(L - \hat{L})c\| \leq \sigma_d(\hat{L}) + \delta$, using Theorem 2 for $\hat{L}$ on S_d and the operator-norm bound. $\square$

Equation (S2) converts one calibration run into a metrological cannot-see guarantee: it is the ingredient separating a numerical null space from a certified instrument property. TLSG arm B measures it directly (Section S4.3).

Theorem 4 (Noncomposability). *For leak routes $L_1, \dots, L_k$ with weights w_i , the simultaneous root-sum-square certificate is governed by the stacked operator $L_{\text{stack}} = [w_1 L_1; \dots; w_k L_k]$, so*

$$\beta_d^2 = \lambda_d^\dagger \left(\sum_i w_i^2 L_i^\dagger L_i \right). \quad (\text{S3})$$

The certificate factors are not multiplicative; multiplication occurs only for serial contractions whose singular-vector chains align.

Proof. $L_{\text{stack}}^\dagger L_{\text{stack}} = \sum_i w_i^2 L_i^\dagger L_i$ by block computation; apply Theorem 2 to L_{stack} . Because the bottom eigenvector of a sum of positive-semidefinite forms is not the product of the individual bottom subspaces, the d -capacities do not multiply. $\square$

Two committed instances of (S3). The two “ $\rightarrow 24$ ” witnesses are distinct and are not conflated. (i) IT5 (@c8a0487): the DPSS hardware route does not stack with the SVD code route—the single-plane certified dimension at 10^{-4} drops $80 \rightarrow 24$, because the window rotates and reshapes the operator. (ii) IT4b (@5666d1e): Tukey apodization is an independent witness of the same law, with joint certified dimension $35 \rightarrow 24$ and a $93\times$ worse relative leak at $d = 64$ (4.66×10^{-2} apodized versus 4.99×10^{-4} non-apodized; a rim effect). These are different operators and different splits.

Extra spectra. The full ascending generalized-leak spectrum σ_d versus d (IT4b, @5666d1e) runs from $\sigma_1 = 1.43 \times 10^{-6}$ through $\sigma_{64} = 4.99 \times 10^{-4}$ to $\sigma_{120} = 1.13 \times 10^{-1}$; the plotted capacity curve of Fig. 1 (center) is a 30-point sampling of this map. Two capacity points anchor it: single-plane $d@10^{-5} = 80$, and joint (four z_1 planes) $d@10^{-4} = 35$, $d@10^{-5} = 8$, $d@10^{-3} = 80$, so the single-plane requirement buys 45 extra certified dimensions. The $d = 64$ subspace has worst-direction leak 4.99×10^{-4} and operational random-code RMS leak 1.68×10^{-4} (threefold below the worst direction), ridge-stable ($d@10^{-4} = 35$ across regularizers $0/10^{-8}/10^{-6}$). Certified-blind codes keep a second sensing channel: eigenvalue ratio $\lambda_{\text{random}}/\lambda_{\text{blind}} = 1.20$, operational mean leak $219\times$ lower (2.58×10^{-4} versus 5.64×10^{-2} ; IT4, @33b8ef2).

S3 Jet transmutation: flow equations and specialization

Let the profiled Chernoff distance carry two leading operators,

$$C(\varepsilon, g) = A g^{2a_1} \varepsilon^{2m_1} + B g^{2a_2} \varepsilon^{2m_2}, \quad 0 < m_1 < m_2, \quad (\text{S4})$$

and set $y = (B/A)g^{2(a_2-a_1)}\varepsilon^{2(m_2-m_1)}$. The effective orders are $m_{\text{eff}} = \frac{1}{2} \partial_{\log \varepsilon} \log C = (m_1 + m_2 y)/(1 + y)$ and $a_{\text{eff}} = \frac{1}{2} \partial_{\log g} \log C = (a_1 + a_2 y)/(1 + y)$, and a direct differentiation gives the exact flow

$$\frac{\partial m_{\text{eff}}}{\partial \log \varepsilon} = 2(m_{\text{eff}} - m_1)(m_2 - m_{\text{eff}}), \quad \frac{\partial m_{\text{eff}}}{\partial \log g} = \frac{2(a_2 - a_1)}{m_2 - m_1} (m_{\text{eff}} - m_1)(m_2 - m_{\text{eff}}), \quad (\text{S5})$$

with fixed points m_1 (relevant as $\varepsilon \rightarrow 0$) and m_2 .

Corollary 1 (Jet transmutation). *A lower-order symmetry-breaking channel with coefficient $A > 0$ drives $m_{\text{eff}} \rightarrow m_1$; zeroing A by an exact wall, a certified code, or a nuisance projection places the experiment exactly on the m_2 manifold without changing the scene perturbation.*

For the near-contact optical case $m_1 = 1, a_1 = 0, m_2 = 2, a_2 = 2$ (covariance Chernoff information is $O(z_2^4 \varepsilon^4)$),

$$\frac{\partial m}{\partial \log \varepsilon} = 2(m - 1)(2 - m), \quad \frac{\partial m}{\partial \log z_2} = 4(m - 1)(2 - m), \quad (\text{S6})$$

so that $y \propto \varepsilon^2 z_2^4$, giving $\varepsilon z_2^2 = \text{const}$ and $z_{2,*} \propto \varepsilon^{-1/2}$, the geometric form of the KT2 crossover $\varepsilon_{\text{cross}} = \sqrt{A/B}$. The measured realizations (KT2, @05f1029): projected covariance slopes $[4, 4, 4, 4]$; unprojected low-amplitude slopes $[2.136, 2.20, 2.117, 2.176]$; orthogonality residuals in $[1.45, 5.09] \times 10^{-18}$; generic-direction covariance slopes $[1.576, 2.284, 1.576, 2.285]$. The cross-paper integer anchors are the companion KL slopes 2.038 (generic) and 4.000 (first-order-orthogonal) with coefficient agreement 1.07%/0.00% (JET_TEST, @1bf29f1; Ref. [COMPANION]).

S4 Three-ledger certifiability: proof and scaling gate

S4.1 Nuisance-profiled LAN reduction

One independent bank yields $Y_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$, $t = 1, \dots, T$. Whiten the null covariance and project every admitted nuisance tangent in the Gaussian covariance metric, giving the efficient displacement $A_\perp = \Pi_\perp [\Sigma_0^{-1/2}(\Sigma_\varepsilon - \Sigma_0)\Sigma_0^{-1/2}]$. Let the efficient symmetric tangent $\mathcal{H}$ have dimension p with basis $B_1, \dots, B_p$ normalized by $\frac{1}{2} \text{tr}(B_j B_k) = \delta_{jk}$, and write $A_\perp = \sum_j \theta_j B_j$, $\lambda = \|\theta\|^2 = \frac{1}{2} \|A_\perp\|_F^2$. The single-bank score $s_j(Y) = \frac{1}{2} [Y^\top B_j Y - \text{tr}(B_j)]$ has $\mathbb{E}_0 s = 0$, $\text{Cov}_0(s) = I_p$, and $\mathbb{E}_{A^\perp} s = \theta + o(\|\theta\|)$, so the normalized aggregate score is locally

$$Z_T = T^{-1/2} \sum_t s(Y_t) \approx \mathcal{N}(\sqrt{T} \theta, I_p). \quad (\text{S7})$$

S4.2 The three ledgers

Ledger I (known direction). For a known unit direction $u = \theta/\|\theta\|$, the matched score $\nu = u^\top Z_T$ has $\nu \sim \mathcal{N}(0, 1)$ under H_0 and $\mathcal{N}(\sqrt{T}\lambda, 1)$ under H_1 , so fixed nontrivial power requires and attains $T\lambda = O(1)$.

Ledger II (blind existence). If only $\|\theta\|^2 = \lambda$ is known, the invariant $Q_T = \|Z_T\|^2$ is χ_p^2 under H_0 (mean p , SD $\sqrt{2p}$) and $\chi_p^2(T\lambda)$ under H_1 (mean $p + T\lambda$), so the norm test separates the hypotheses when $T\lambda \gg \sqrt{p}$.

Lemma 2 (Minimax lower bound). *Mixing the alternative uniformly over directions of norm $r = \sqrt{T}\lambda$, the χ^2 -divergence of the mixture from the null satisfies $\chi^2 + 1 = \mathbb{E} \exp(\mu^\top \mu')$ for independent mixture directions μ, μ' , with $\mathbb{E}(\mu^\top \mu') = 0$ and $\text{Var}(\mu^\top \mu') = r^4/p$. If $r^4/p = T^2 \lambda^2/p \rightarrow 0$ the divergence tends to zero, hence total variation tends to zero and every test has asymptotic error sum approaching one. No direction-agnostic procedure succeeds below $T\lambda \sim \sqrt{p}$; the norm test attains this scale.*

Ledger III (estimate the direction). The efficient estimator $\hat{\theta} = Z_T/\sqrt{T}$ has exact local risk $\mathbb{E}\|\hat{\theta} - \theta\|^2 = p/T$, matching the Cramér–Rao bound, so relative squared error below a constant requires $T\lambda \sim p$. Hence

$$\text{known} : \text{blind} : \text{full} = 1 : \sqrt{p} : p. \quad (\text{S8})$$

Composing with the physical jet $A_\perp = \kappa g^a \varepsilon^m B$ ($\frac{1}{2} \|B\|_F^2 = 1$) gives $\lambda = \kappa^2 g^{2a} \varepsilon^{2m}$ and $T_{\text{known}} \asymp \kappa^{-2} g^{-2a} \varepsilon^{-2m}$, $T_{\text{blind}} \asymp \sqrt{p} T_{\text{known}}$, $T_{\text{est}} \asymp p T_{\text{known}}$; adaptation multiplies the prefactor and leaves the physical exponent $2m$ unchanged. Boundary cases: at $p = 1$ the three ledgers coincide; an exact wall $B = 0$ gives $\lambda = 0$ and no finite T succeeds; a sparse admitted class replaces p by its metric entropy.

S4.3 Frozen scaling gate (TLSG)

The gate is the falsification experiment above plus a calibration-transfer arm for Eq. (S2). Arm A sweeps $M \in \{8, 16, 32, 64\}$ and $p_{\text{eff}} \in \{10, 36, 136, 528\}$ (eight (M, p) cells), $\lambda = 0.15$, $N_{\text{MC}} = 1500$, false-positive rate 0.05, seed 20260724. Arm B uses $d_{\text{cert}} = 64$, $n_{\text{cal}} = n_{\text{test}} = 24$, conjugate $z_1 = 0$, and perturbations fill $\pm 5\%$, pitch $\pm 3\%$, relay defocus 0–0.5 mm. All seven bars pass; the verdict is TLSG_PASS at commit 35d858e (Table S2).

Bar 4 is the $1 : \sqrt{p} : p$ signature; bar 3’s coefficient 0.985 is the Cramér–Rao floor p/T ; bar 5 confirms the KT6 negative for the right reason (the operating point lies below the optimal blind tier). The oracle law itself is intact: the near-contact ratio $J_{\text{cov}}/J_{\text{mean}} \propto z_2^4$ has fitted exponents $[4.081, 4.054, 4.022, 3.989]$, but the blind median signal-to-floor ratio is $\approx 5.4 \times 10^{-3}$ (worst 5.25×10^{-3}), so the estimator is killed (KT6, @635ff05).

Table S2. TLSG gate: all seven frozen bars (TLSG_RESULTS.json, @35d858e). Values are quoted verbatim from the committed artifact.

#	Bar (frozen target)	Measured	Verdict
1	Matched-score 50%-power contours collapse vs $T\lambda$ (CV ≤ 0.20)	CV = 0.1250	pass
2	Blind contours collapse vs $T\lambda/\sqrt{p}$ (CV ≤ 0.25) AND fail vs $T\lambda$	CV _{scaled} = 0.1242; CV _{raw} = 0.6283	pass
3	Relative estimation risk collapses vs $T\lambda/p$ (CV ≤ 0.25)	CV = 0.0348 (mean risk $\cdot T\lambda/p = 0.985$)	pass
4	Fitted p -exponents (matched/blind/estimation): $0/\frac{1}{2}/1 \pm 0.15$	-0.004 / 0.444 / 1.010	pass
5	KT6 operating point below blind power 0.2 at FPR 0.05	blind power = 0.0506 ($10 \times = 0.0560$, $100 \times = 0.1335$; $p=136$, $T\lambda=0.0866$)	pass
6	Fresh calibration/test: $\geq 95\%$ simultaneous coverage of $\sigma_d(\hat{L}) + \delta$	coverage = 1.000 ($\sigma_{64}=3.41 \times 10^{-8}$, $\delta=1.07 \times 10^{-2}$; worst fresh S_{64} leak 1.99×10^{-6})	pass
7	Conjugate plane: ≥ 64 code dims certified $< 10^{-4}$ on fresh law draws	fresh $d@10^{-4}$ min = 80 (mean 80.0; calibration $d@10^{-4} = d@10^{-5} = 80$)	pass

S5 Theorem-consistent honest negatives

The kills carry the same prominence as the positives. Two apparatus shortcuts were falsified under preregistered protocols and are recorded here in full.

Segmented-bucket negative (IT6, @dade877). The P^2 information law is real (fitted exponent 1.68–2.02; $P = 4$ buys 12–17 $\times$), but segmenting the bucket *breaks the mean wall*: the per-segment discriminability d' grows from 0.8 to 1.8 at $z_2 = 5$, so beyond-band change floods the segmented mean channel. The single, whole-aperture bucket stands; segmentation is not a valid capacity multiplier. Verdict: KILL.

Coincidence-veto negative (IT7, @e1546ac). A contact-arm coincidence veto is a valid medium-event discriminator—the contact arm is exactly medium-blind (AUC = 0.5009) and the medium false-alarm rate is repaired 17.8 $\times$ (from 0.9965 to 0.056)—but it is an invalid beyond-band repair, retaining only 24% of scene power, because the same wall that supplies specificity starves the contact arm of beyond-band signal. It belongs in the bench manual as a theorem-consistent negative control, not as a method. Verdict: KILL. This is why no strong-specificity wording is claimed, and the semiparametric projection remains a quarantined future method until estimator-layer certification is solved.

Future objects (deferred, not claimed). A task-dependent étendue law $N_* = [\nu/(2 - \nu)]n$ is only partially collapsed (KT5, @752114a: N_{eff} grows 46 $\rightarrow$ 811, normalized CV 0.64), and coherence-rank unsaturation persists with a thin screen holding $R \approx 110$ without an $R \rightarrow 1$ collapse (KT4, @494b46b). Both are gated future work (named gates étendue-matching and COHERENCE-CROSSOVER); neither supports a claim here. The memory-effect phase diagram requires a genuine thickness/stacking axis and is deferred.

S6 Model and protocol capsule

Engines and shared configuration. The wall/jet exams use the exact Kronecker speckle generator (fully developed circular-complex-Gaussian field, factorized grain $\otimes$ code covariance), grid 2048 for IT1/KT1 in `complex128/float64`, with code counts $M \in \{8, 12, 16, 32\}$ and 200–700 banks per test. The blindness-capacity and route tests use the SVD/DPSS leak operators at the relay-conjugate plane. The COMSOL full-Maxwell twin (Wave Optics, 2D TE, COMSOL 6.3) validates the thin-phase screen at 15.5% contrast agreement (0.467 versus 0.395; WAVE_TWIN, @0c3b0d9). No reconstruction is performed anywhere.

Table S3. Frozen-source provenance (commit at freeze). Full field-level map in `paper2-optica/CLAIM.SOURCE.MATRIX.md`.

Artifact	Commit	Role
TLSG_RESULTS.json	35d858e	three-ledger scaling + calibration transfer (gate)
KT1 shell sweep	19338a4	support/field wall 6.6×10^{-16} + breakers
IT1 two-wall jet	f13a323	energy/unitary wall 2.18×10^{-16} ; NA-clip ε^1
KT2 jet transmutation	05f1029	projected slope 4.00; jet flow
KT3 orthogonalized sentinel	2430166	code-space projection, 86–99% retained
IT4 / IT4b	33b8ef2 / 5666d1e	capacity spectrum (80/35/24 dims)
IT5 DPSS concentration	c8a0487	DPSS route 2.4×10^{-4} ; $80 \rightarrow 24$
KT6 channel-ratio ranging	635ff05	third-ledger boundary (oracle z_2^4)
IT2 / IT3 / IT8	acfcda0 / 6fb5149 / 6fb5149	floor decomposition, leak law, PWM
IT6 / IT7	dade877 / e1546ac	honest negatives (Section S5)
WAVE_TWIN	0c3b0d9	COMSOL 15.5%, $M^{1.8}$, matched cell $768 \in [513, 2996]$
JET_TEST / SCRAM- BLE_EXT	1bf29f1 / ed7a1e0	companion jet/scrambling anchors

Scaling gate. Arm A and arm B parameters are in Section S4.3; the single seed is 20260724; there was one sealed run and no repair round.

Provenance. Every number in the article and this Supplement traces to one committed frozen artifact (Table S3); the extracted verbatim copies read by the figure scripts live in `figures/_frozen_sources/`. The working tree is clean for all cited artifact directories.

Data availability. The committed artifacts, claim–source matrix, and figure scripts are available at [URL].

References

- [1] [AUTHORS], Strong optical disorder erases images but preserves local change observability, companion Letter (2026).